AutoML

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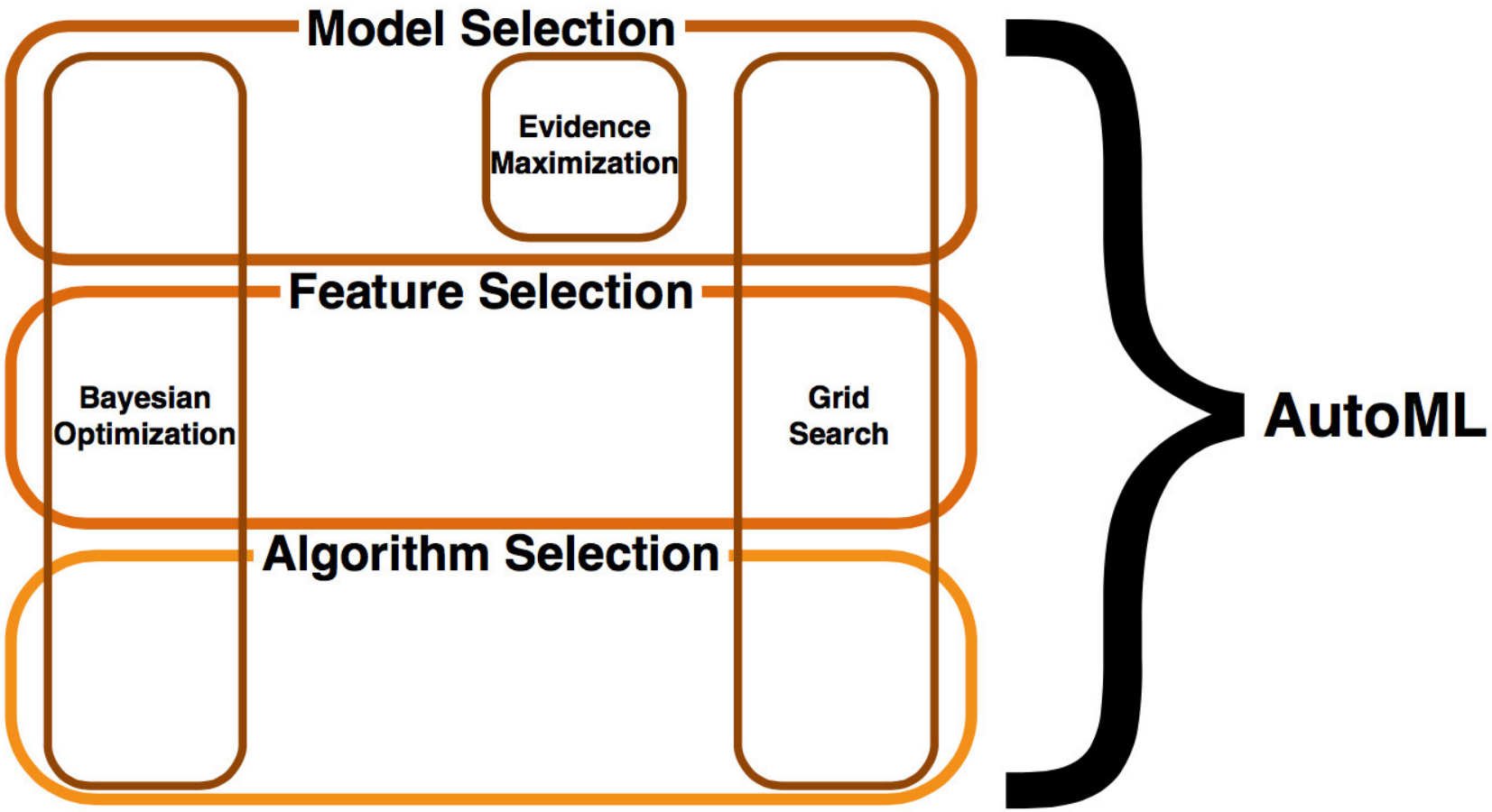
Big Picture of AutoML



Common Model Parameter Learning

- A model parameter learning is

$$\underset{\theta \in \Theta}{\operatorname{argmin}} \, \mathcal{L}(\theta; \lambda_i, A_i, F_j, \{\mathbf{x}_k, y_k\}_{k=1}^{n})$$

where $\mathcal{L}$ is a loss function, f is a predictive model, θ is a parameter vector, λ_i is the hyperparameter vector of the chosen algorithm A_i, and F_j is the chosen feature vector. $\mathbf{x}_k$ is an input value and y_k is a output value.

AutoML

- Each problem optimizes parameter, hyperparameter, feature and model.
  - Algorithm, feature and hyperparameter selection

$$\underset{\theta\in\Theta,\lambda_i\in\Lambda,F_j\in\mathcal{F},A_i\in\mathcal{A}}{\operatorname{argmin}} \mathcal{L}(\theta,\lambda_i,F_j,A_i;\{\mathbf{x}_k,y_k\}_{k=1}^n)$$

  - Feature and hyperparameter selection

$$\underset{\theta\in\Theta,\lambda_i\in\Lambda,F_j\in\mathcal{F}}{\operatorname{argmin}} \mathcal{L}(\theta,\lambda_i,F_j;A_i,\{\mathbf{x}_k,y_k\}_{k=1}^n)$$

  - Model selection

$$\underset{\theta\in\Theta,\lambda_i\in\Lambda}{\operatorname{argmin}} \mathcal{L}(\theta,\lambda_i;F_j,A_i,\{\mathbf{x}_k,y_k\}_{k=1}^n)$$

Bayesian Optimization

- A powerful strategy for finding the extrema of objective functions that are expensive to evaluate,
  - where one does not have a closed-form expression for the objective function, but where one can obtain observations at sampled values.
- To incorporate prior belief to help direct the sampling and to trade off exploration and exploitation of the search space.
- The prior represents our belief about the space of possible objective functions.

[1] E. Brochu, V. M. Cora, N. de Freitas. A tutorial on Bayesian optimization of expensive cost functions, with application to active user modeling and hierarchical reinforcement learning. Technical Report UBC TR-2009-23 and arXiv: 1012.2599v1, 2009.

Gaussian Process

- GP is expressed as $f(x) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$.
- The squared-exponential covariance functions in one dimension has the following form,

$$k_y(\mathbf{x}_p, \mathbf{x}_q) = \sigma_f^2 \exp(-\frac{1}{2l^2}(\mathbf{x}_p - \mathbf{x}_q)^2) + \sigma_n^2 \delta_{pq},$$

  where σ_f is the signal standard deviation, l is the length scale and σ_n is the noise standard deviation.
- The mean and covariance of the posterior are

$$Mean = k_{xx'} k_{x'x'}^{-1} y,$$
$$Covariance = k_{xx} - k_{xx'} k_{x'x'}^{-1} k_{x'x}.$$

[2] C. E. Rasmussen and C. K.I. Williams. Gaussian Processes for Machine Learning. MIT Press, 2006.

Gaussian Process

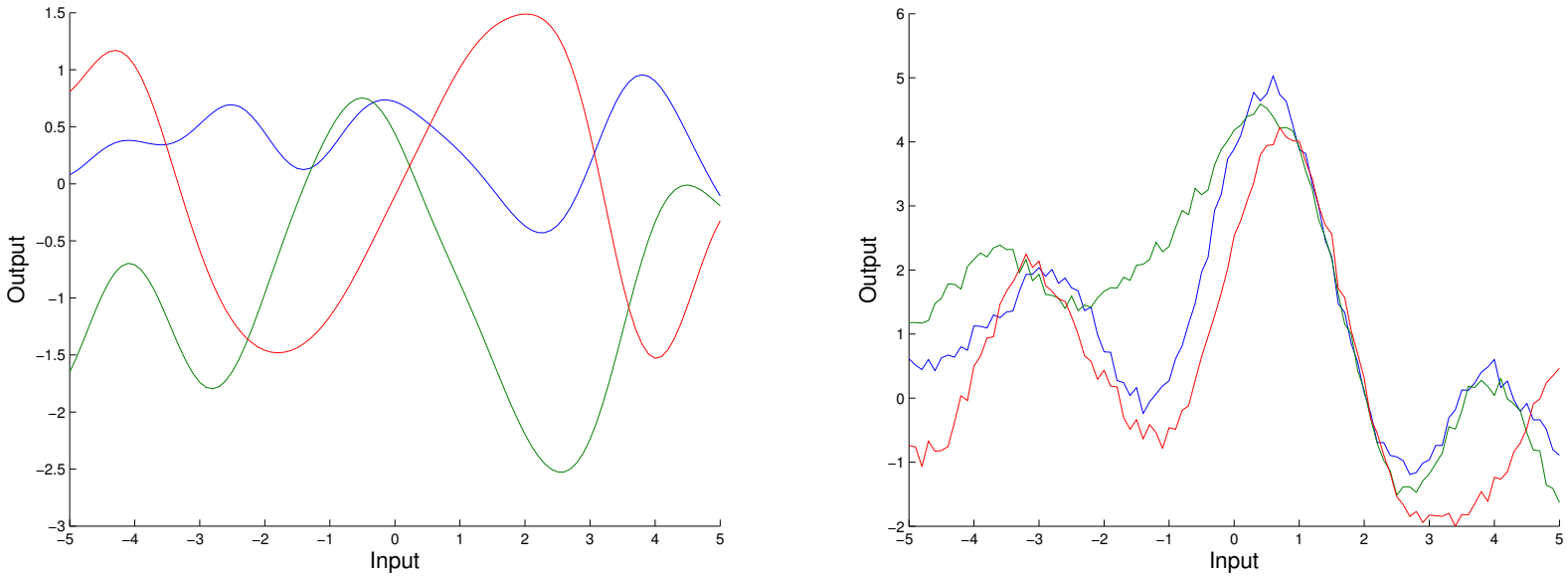


Figure: GP prior (left) and posterior, given three points, with noise (right). Length scale = 1.0, signal standard deviation = 1.0 and noise standard deviation = 0.1.

Acquisition Function for Bayesian Optimization

- Probability of Improvement (PI)

$$a_{PI}(\mathbf{x};\{\mathbf{x}_n, y_n\},\theta) = \Phi(\gamma(\mathbf{x})),\ \gamma(\mathbf{x}) = \frac{\mu(\mathbf{x};\{\mathbf{x}_n, y_n\},\theta) - f(\mathbf{x}_{best})}{\sigma(\mathbf{x};\{\mathbf{x}_n, y_n\},\theta)}$$

- Expected Improvement (EI)

$$a_{EI}(\mathbf{x};\{\mathbf{x}_n, y_n\},\theta) = \sigma(\mathbf{x};\{\mathbf{x}_n, y_n\},\theta)(\gamma(\mathbf{x})\Phi(\gamma(\mathbf{x})) + \mathcal{N}(\gamma(\mathbf{x});0,1))$$

- GP Upper Confidence Bound (GP-UCB)

$$a_{UCB}(\mathbf{x};\{\mathbf{x}_n, y_n\},\theta) = \mu(\mathbf{x},\{\mathbf{x}_n, y_n\},\theta) + \kappa\sigma(\mathbf{x};\{\mathbf{x}_n, y_n\},\theta)$$

[1] E. Brochu, V. M. Cora, N. de Freitas. A tutorial on Bayesian optimization of expensive cost functions, with application to active user modeling and hierarchical reinforcement learning. Technical Report UBC TR-2009-23 and arXiv: 1012.2599v1, 2009.

Gaussian Process Posterior and Acquisition Function

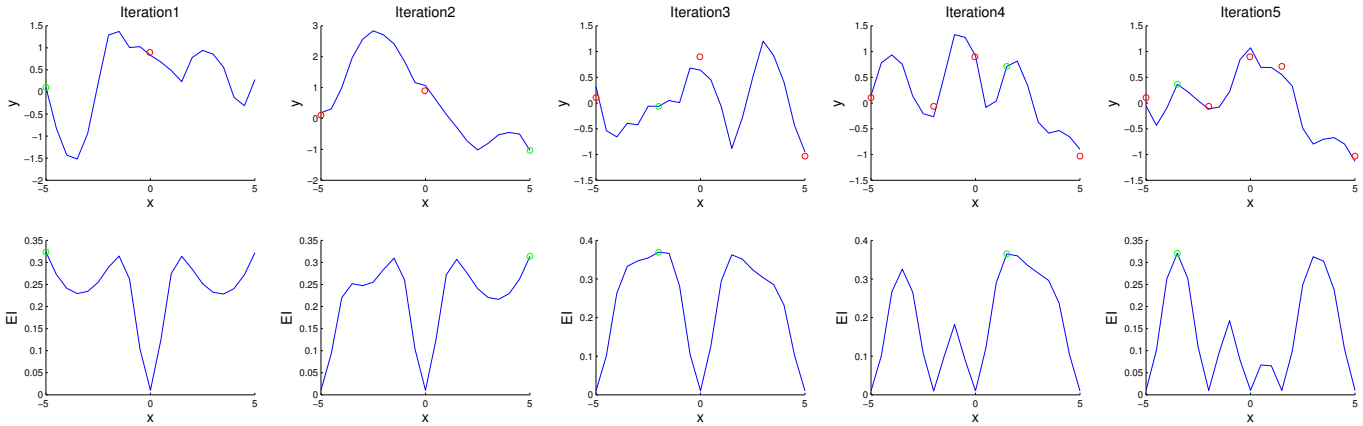
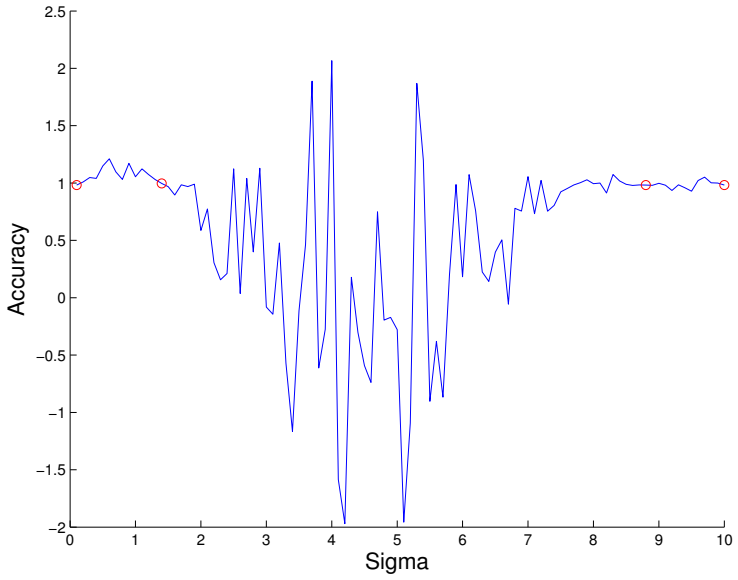


Figure: 5 iterations to apply EI criterion. Length scale = 1.0, signal standard deviation = 1.0 and noise standard deviation = 0.1.

Bayesian Optimization for SVM



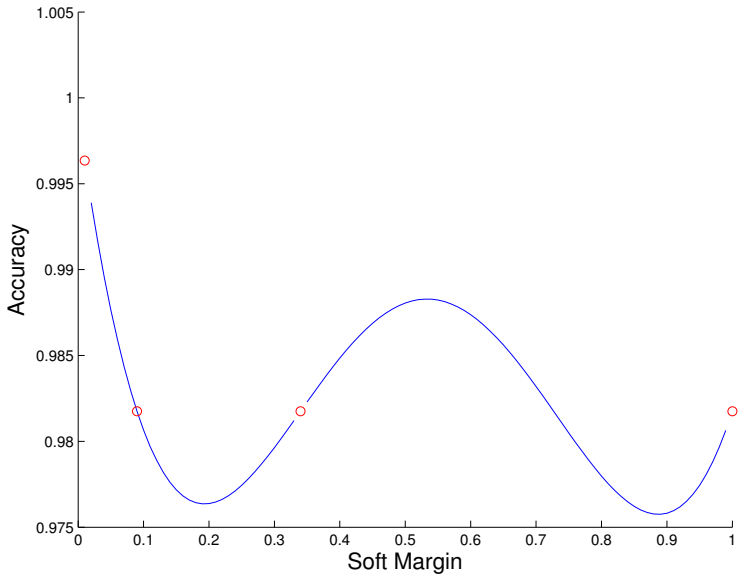


Figure: The result applied BO with GP for SVM hyperparameters, s and soft margin. It iterates 5 times to get the point acquired by EI criterion. Data is from banknote authentication data set of UCI Repository. Length scale = 1.0, signal standard deviation = 1.0 and noise standard deviation = 0.0.

Bayesian Optimization for SVM

- The hyperparameters is optimized to

$$s = 4.0,$$
$$soft\ margin = 0.02.$$

- The accuracy predicted by the optimized hyperparameters of SVM is 0.9927.
- The accuracy predicted by MATLAB default SVM function is 0.9854.

Reference

[1] E. Brochu, V. M. Cora, N. de Freitas. A tutorial on Bayesian optimization of expensive cost functions, with application to active user modeling and hierarchical reinforcement learning. Technical Report UBC TR-2009-23 and arXiv: 1012.2599v1, 2009.

[2] C. E. Rasmussen and C. K.I. Williams. Gaussian Processes for Machine Learning. MIT Press, 2006.